

MRI-guided Breast Biopsy Case-based Review: Essential Techniques and Approaches to Challenging Cases

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Abbreviation: ACR = American College of Radiology

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Breast MRI is an established imaging modality for screening and diagnosing breast cancer. MRI-guided biopsy helps determine the histopathologic benignity or malignancy of suspicious MRI findings without conclusive correlative findings at other imaging modalities such as mammography or breast US.

The American College of Radiology (ACR) accreditation process requires that medical facilities offering breast MRI services perform imaging correlation with mammography, MRI-directed (second-look) breast US, and MRI-guided interventional procedures, or have a less formal or contractual referral partnership with a medical institution that offers these services. The ACR Committee on Breast MRI Accreditation requires breast MRI biopsy capability, in-house or by a partner facility, to ensure patient safety and reduce the patient's expenses. The ACR does not dictate the nature of the referral partnership but strongly recommends that the referral facility have breast MRI accreditation by the ACR. Successful MRI-guided biopsies require skilled experienced radiologists and technologists who are familiar with the biopsy techniques and can problem solve when faced with cases that require additional prebiopsy planning.

A meticulous review of the diagnostic MR images to establish the quadrant, clock position, depth, distance from the nipple, and the two-view visualization is performed to determine the best approach for the biopsy. MRI-guided breast biopsies are limited to lateral-to-medial or medial-to-lateral approaches.

Optimal positioning of the patient by using a team-based approach is one key factor for biopsy success. First, the affected breast is compressed with the help of a compression device with a sterile grid. Typically, the patient's arms are positioned above the head. The amount of compression should not be so tight as to impede enhancement, but the skin should be taut to prevent motion and stabilize the breast for adequate sampling.

Radiologic-pathologic concordance is a key part of image-guided biopsy procedures, whether the radiologist deems the target lesion simple or challenging to sample. Concordance becomes more critical with MRI-guided breast biopsies, as there are inherent uncertainties with the accuracy of sampling. Specimen radiography is not available with MRI-guided biopsies as in the case of stereotactic-guided biopsies, nor is the biopsy needle monitored in real time as with US-guided biopsies.

Concordance starts before the first sample is taken (Fig 1). The radiologist should make an initial assessment about the expected pathologic results when he or she reviews the original breast MR images. There is radiologic-pathologic discordance when the radiologist determines that the pathologic result does not agree with the initial assessment (Fig 2). Discordant results should lead to either surgical excision or rebiopsy.

TEACHING POINTS

- In cases where there is a suspicious MRI finding and no conclusive correlative finding at other imaging modalities such as mammography or breast US, MRI-guided biopsy is necessary for the histopathologic determination of benignity or malignancy.
- Meticulous prebiopsy planning is an important part of the breast MRI-guided biopsy process.
- Radiologic-pathologic concordance starts when the radiologist makes an initial assessment about the expected pathologic results after reviewing the original breast MR images. If the results are discordant, then the suspicious finding should be rebiopsied or surgically excised.

The importance of MRI-guided biopsy technique and problem solving cannot be overstated. In this online presentation, we review the indications and contraindications for MRI-guided breast biopsies; detail the prebiopsy planning process, including equipment and biopsy techniques; illustrate tips, tricks, and troubleshooting methods; and highlight the importance of radiologic-pathologic correlation.

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Suggested Readings

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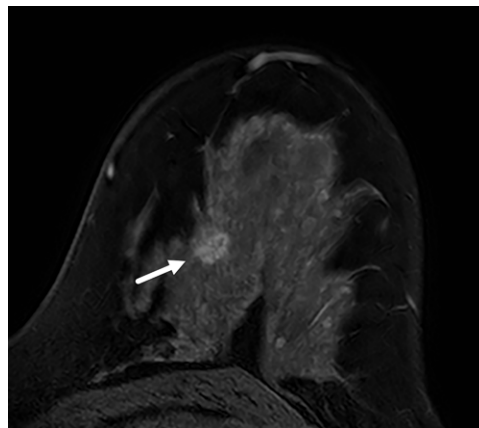


Figure 1. Axial postcontrast T1-weighted MR image shows an irregular enhancing mass (arrow) in the left breast. The mass was not seen at MRI-directed US. Therefore, MRI-guided biopsy was recommended.

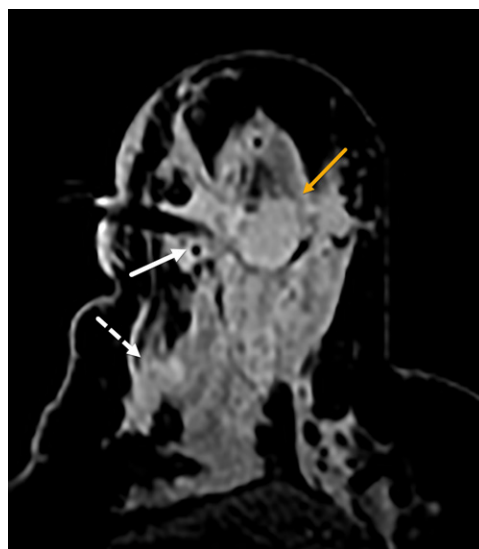


Figure 2. Axial MR image from an MRI-guided biopsy that was performed at an outside facility shows the tip of the needle (white solid arrow) with an associated postbiopsy hematoma (yellow arrow) on postsampling sequences. Pathologic analysis showed atypical lobular hyperplasia. Initially, the radiologist deemed the pathologic results concordant with imaging. However, careful review of the biopsy and prebiopsy images showed that the biopsy target is more posterior (white dashed arrow). Repeat MRI-guided biopsy was recommended. Pathologic analysis showed invasive ductal carcinoma. The complete case is available in the online presentation.